Risk Management and Healthcare: IoT Technologies and Smart Monitoring System for a Good Cold Chain Management

Teresa Bengiovanni
Dept. of Technological Innovation
Local Health Authority of Matera
Matera, Italy
teresa.bengiovanni@asmbasilicata.it

Roberto Conte Rosito
Software Architect
I.T.S. S.r.l.
Matera, Italy
roberto.conterosito@itsmt.it

Vincenzo Lacasa
CEO
I.T.S. S.r.l.
Matera, Italy
vincenzo.lacasa@itsmt.it

Eugenio Simone
Dept. of Technological Innovation
Local Health Authority of Matera
Matera, Italy
eugenio.simone@asmbasilicata.it

Ilaria Sergi
Dept. of Innovation Engineering
University of Salento
Lecce, Italy
ilaria.sergi@unisalento.it

Vincenzo Iacovone
Full Stack Developer
I.T.S. S.r.l.
Matera, Italy
vincenzo.iacovone@itsmt.it

Michele Viggiano
Department Manager of Local Health Authority of Matera
Matera, Italy
mviggiano@asmbasilicata.it

Luigi Patrono
Dept. of Innovation Engineering
University of Salento
Lecce, Italy
luigi.patrono@unisalento.it

***Abstract*— The management of clinical risk in the health sector is mandatory in order to guarantee the safety of patients and operators. In this regard, several regulations and guidelines have been developed by national and international organizations. One of the most critical issues, currently not fully managed, is the safe transportation of temperature sensitive medicines, vaccines, and biological samples. Thanks to the diffusion of emerging technologies enabling the Internet of Things such as embedded systems, mobile Apps and Cloud services, it is possible to obtain continuous monitoring of sensitive substances and environments. In these contexts, particular environmental parameters (e.g., temperature and humidity) can be processed in real time to promptly trigger specific extraordinary and ordinary maintenance interventions. In the present work, the aforementioned technologies have been exploited in order to develop an innovative monitoring of refrigerated storage systems. The proposed solution has validated from a functional point of view through a real use case.**

Keywords—Internet of Things, Risk management, Good Cold Chain Management, Smart Environments, Monitoring, REST.

I. Introduction

Health risk management is very complex due to the variety of clinical pathways and related risks. An effective risk management system is needed to ensure that each clinical pathway takes place safely. To achieve this goal, once processes and critical elements have been identified and analysed, it is necessary to identify tools and solutions to guarantee risk reduction: implementation of correct procedures and actions, use of training tools, appropriate finding and selection of technical/technological solutions that can reduce risk, by simplifying its management and ensuring the traceability of the related activities. To meet these needs, it is possible to benefit from the recent advantages in Internet of Things (IoT). The improvements in Wireless Sensor Networks (WSNs) energy management [1, 2, 3] and the spread of embedded devices, Cloud services, sensors, actuators and other low-cost and low-power devices offers a great opportunity to realize smart solutions capable of simplifying and improving the health risk management. In this perspective, a hot topic concerns the so-called Cold Chain Management [4] which also represents the theme of the proposed work.

In particular, in this paper, a system for monitoring the refrigerated storage of drugs, vaccines, medical devices and any biological samples (blood and its derivatives, saliva, urine, cells of various types and classification) is presented. The proposed system offers immediate alarm tools in case of malfunction of the refrigeration systems (fridge and/or freezers). It is also suitable for use in any sector that requires the maintenance and control of physical parameters (e.g., temperature and humidity) to ensure the correct certification of processes and the containment of economic damages deriving from the malfunctioning of the refrigeration systems. In the context of risk management in the health sector, the main objective of this work is to certify the internal and external environmental parameters of biological refrigerators and freezers in order to preserve the characteristics of the transported samples and their effectiveness until the expiration date (shelf life, in case of drugs, vaccines and medical devices). Strongly related is both the health and legal aspect: such a system, if properly implemented, it allows to safeguard the stored samples, thus avoiding the deterioration risk and their loss, especially if unique and not replicable. In addition, it simplifies the activities of both health workers, responsible for temperature control, and maintenance operators, responsible for the proper functioning of the refrigeration systems. Finally, this system also allows a reduction of personnel costs as well as a reduction of costs for the disposal of drugs, vaccines and medical devices in the event of a failure of the biological fridge/freezer storing them. For the development of the proposed system, the regulations and guidelines in force in Europe for the correct refrigerated

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on January 07,2026 at 09:45:34 UTC from IEEE Xplore. Restrictions apply.

storage of drugs, vaccines, medical devices and biological samples have been considered:

- Drugs: CPMP/QWP/609/96 guidelines "Declaration of Storage Conditions for Medicinal Products Particulars and Active Substances", (BPD/GDP) 2013/C 343/01 guideline "Good distribution practices for medicines for human use, 5/11/2013 [5];
- Tissue and cells: directives 2006/17/CE, 2006/86/CE, 2004/23/CE, guidelines "Guide to the quality and safety of Tissues and Cells for human application, EDQM (2nd edition 2015)" [6];
- Blood, components and derivatives: directive 2016/1214/CE, 2005/62/CE, guideline "Guide to the Preparation, Use and Quality Assurance of Blood Components (19th Edition)" [7];
- Medical devices: European regulation (EU) 2017/745 [8].

The characteristics of some commercial refrigeration systems have been analysed from a technical point of view. Traditional systems making use of graphs recorders, diskettes, pens for temperature traceability and electrical contacts for remote alarm controls have not been taken into account. Refrigerator system's producers require temperature storage systems (datalogger, SD card, etc.), but data transfer is mainly manual (via USB port or the classic RS 485/422/232), as noted in the datasheets of the main Italian manufacturers. Alarm recording and transmission systems are also available if a malfunction or a deviation from the configured ranges occurs. But the communication protocols available in these systems (Modbus, TCP IP) do not always make the stored data immediately and fully usable. It is possible to find devices equipped with a LAN, UMTS/3G, Wi-Fi, or GSM interface, as noted in the datasheets of the main Italian manufacturers, but often these additional interfaces represent an option that significantly affects the final price, especially in the case of small refrigeration systems. There are also several software for managing the data collected, but these are proprietary programs that do not allow managing the traceability of refrigeration systems of other manufacturers, as noted in the datasheets of the main Italian manufacturers. Finally, there are mature thermologger solutions but, in the context of Industry 4.0, it is no longer possible to conceive devices that are not interconnected on the network or not capable of capturing and processing alerts to support intervention and maintenance systems. Despite the many solutions on the market, there is no a simple and low-cost solution suitable for all temperatures and volumes typical of the healthcare environment (analysis laboratories, immunotransfusion services, medical genetics laboratories, tissue typing laboratories for transplants, pathological anatomies , cryopreservation laboratories, hospital and regional pharmacies, hospital operating units, territorial clinics). The solution proposed in this work, called SensIC, aims to meet these needs.

The proposed system aims to propose a simple and low-cost hardware and software solution designed for monitoring refrigerated storage systems (refrigerators, freezers, freezer cabinets) in the healthcare sector and for intelligent risk management if a malfunction occurs. A proof-of-concept is also reported which shows the main system functionalities available to all healthcare professionals, with a particular focus on both risk managers and clinical engineers.

This paper is organized as follows. In Section II, main system requirements are presented and in Section III the system architecture is described. A functional validation of the developed system is proposed in Section IV whereas a discussion about the benefits in the use on this solution is reported in Section V. Finally, conclusions are drawn in Section VI.

II. System Requirements

The SensIC system consists of:

- a smart device, called SensBox;

- a Cloud infrastructure dedicated to device management.

In the design of the smart device, the main need was to create a low-cost but high-tech device. SensBox is the result of a careful choice of hardware components. It is based on the Discovery Kit for IoT Node [9] by STMicroElettronics equipped with an STM32L4 microcontroller [10].

The SensBox is responsible for data collection and forwarding and its microprocessor is designed to operate at low energy consumption. In addition to Bluetooth Low Energy (BLE) and Sub-GHz (868MHz), it also integrates a Wi-Fi interface (ISM43362) which can be used, in place of the Ethernet interface, for data forwarding. The board also offers support for Arduino Uno connectivity and other STMicroElettronics components providing unlimited expansion capabilities with a large choice of specialized add-on boards.

The standard equipment includes expansion pins to mount external probes. In particular, analog and digital inputs are reserved for external sensors. Currently, SensBox can be equipped with:

- six DS18B20 probes [11]. With cable length from 1 to 10 meters, this sensor is able to detect temperatures with precision from 9 to 12 bits in a temperature range between -55 ° C and +125 ° C with an approximation of +/- 0.5 ° C. The peculiarity of this probe is the unique 64-bit serial number. This feature allows the use of a large number of sensors on a single data BUS.
- two Pt100 probes [12]. With cable length from 1 to 10 meters, this sensor is a platinum resistance thermometer with a nominal resistance defined according to IEC 751 (EN 60751) equal to 100 Ω at a temperature of 0 ° C. These probes have been used for many years in industrial research environments (laboratories). Compared to thermoelements, the use of a platinum resistance thermometer has several advantages, i.e. wider temperature range (from -200 ° C to +850 ° C), almost linear characteristic curve, excellent precision, stability and long life.

SensBox can be powered via Power over Ethernet (PoE). It is also equipped with an external battery that guarantees its operation for several hours if an electrical blackout occurs. Moreover, an internal memory is used to store measurements in the presence of a lack of connectivity. The device communicates continuously with the Cloud platform. For this reason, security mechanisms must be provided to preserve the recorded data and forward them later in case of technical problems. The firmware developed for the management of the hardware components, in addition to performing orchestration activities (verification of connectivity, switches between the

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on January 07,2026 at 09:45:34 UTC from IEEE Xplore. Restrictions apply.

various types of power supply, recognition of hot-connected sensors and backups), is able to interact with the Cloud platform and adapt to it according to some directives. Specifically, this feature, called *On-Alert*, allows the device to modify its schedules relating the data reading and forwarding in case of emergency situations. Technically, at each measurement communicated to the Cloud infrastructure, the same responds with a coded message, which indicates the possible presence of anomalous situations (e.g. too high temperature). In these cases, the device changes some internal parameters by modifying, for example, the time intervals used to forward the measurements to the Cloud platform. For example, a device that generally forwards data every 5 minutes, if an anomaly is detected, starts the *On-Alert* mode and forwards data every 60 seconds to better monitor the detected anomaly.

As already highlighted, SensBox is in constant communication with the Cloud infrastructure that must be correctly calibrated based on the needs of this type of IoT infrastructure. In this perspective, it is essential that the IT tools are able to quickly manage a large amount of data. In fact, a single device, within 24 hours and at full load (i.e. with all the internal probes active and all the digital and analogue ports connected) is able to produce from 3,000 to 10,000 telemetry measures.

The SensIC platform is able to guarantee a response time less than 0.5s for each communication and, at the same time, extremely fast access to all consultation and data monitoring interfaces.

The SensIC platform offers three types of interfaces accessible to the end user: smartphone application, mobile website, desktop website.

A mobile application for Android smartphone has been designed for the installation and configuration of each SensBox. This application exploits the Bluetooth technology to easily configure the SensBox allowing the management of different parameters:

- Ethernet/Wi-Fi configuration parameters;
- Storage intervals;
- Alarm thresholds;
- Backup intervals if blackout occurs.

A Single Page Application (SPA) for desktop/mobile devices has been developed. It offers all system functionalities and is optimized to work correctly on both smartphones, computer desktops and tablets. The functionalities of the Web application can be grouped in:

- Dashboards: functionalities accessible to end-users;
- Configuration: functionalities accessible to technicians and system administrators;
- Models: functionalities accessible to technicians, engineers and administrators that allow the design of new SensBox typologies (in terms of on-board sensors)
- Administration: functionalities accessible to administrators for the management of users, clients as well as the overall system.

III. System Architecture

The system architecture is presented in Fig.1.

Several technologies have been exploited in the design and development of the SensIC infrastructure.

The PHP language has been chosen for the development of all server-side components.

For data management MariaDB [13] has been used. It is a database engine that inherits the main features of MySQL. Compared to MySQL, MariaDB offers better performance in managing a large amount of data.

All system REST APIs have been realized through the CakePHP framework [14]. It is an advanced product, based on the PHP language, which allows a quick and fast development of applications or parts of web applications.

To realize the hardware infrastructure, Hetzner company [15] has been chosen and its Cloud infrastructure has been selected for the application deployment. It guarantees low-cost, scalability and progressively higher performance based on the system workload. The configured virtual machines have the following equipment:

- Debian/Ubuntu operating system
- Nginx as a proxy server in association with FastCGI Process Manager (PHP-FPM) for processing requests.

The use of Debian/Ubuntu in association with Nginx and PHP-FPM allows to obtain further benefits compared to the use of Apache (generally used for hosting PHP applications). Main benefits are:

- advanced process management (possibility to stop and restart single instances of threads);
- ability to start multiple processes, that is, multiple threads to manage multiple system access queues;
- advanced logging for warning and error diagnostics.

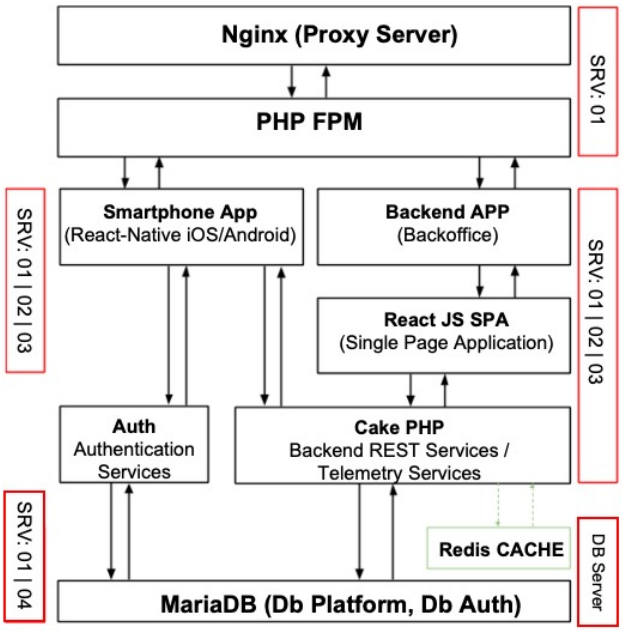


Fig. 1. SensIC architecture.

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on January 07,2026 at 09:45:34 UTC from IEEE Xplore. Restrictions apply.

SensIC is an extremely complex platform, therefore a precise chain has also been designed to guarantee Continuous Integration and Continuous Deployment activities.

Particular attention has been paid to the development of graphical interfaces. SensIC is a real-time monitoring platform based on the display of historical data series as well as telemetry values received in real time via web sockets. To meet these requirements, all front-end interfaces have been created with ReactJS [16]. The Redux architectural pattern has been associated to support real-time monitoring.

The choice of ReactJS as a library for the development of the front-end is motivated by its following characteristics:

- *Easy integration*. ReactJS is a library and not a framework dedicated to the development of web applications. Then, it is possible to integrate React also on existing projects.

- *Many libraries and a large community*. ReactJS offers additional modules and many ready-to-use libraries to face the most common problems and needs. React has a large and active community of developers.

- *Virtual DOM*. It is an abstraction in memory of the components of the User Interface. It offers efficient change management in the DOM thanks to the selective generation of DOM sub-trees at each change of state.

- *JSX*. It is an optional markup language similar to HTML. It allows the declarative description of the interface by combining the ease of writing the Views with the power of the JavaScript language.

- *One-way data binding*. It avoids the complexity of two-way data binding by defining a one-way data flow and a precise communication hierarchy.

- *Code sharing between client and server*. Unlike other client-side libraries, ReactJS is an isomorphic library. This allows the rendering of pages also on the server side.

The main difference among SensIC and other similar solutions is how telemetry data received by each SensBox are processed and managed. Fig. 2 shows a diagram summarizing the main activities carried out by the monitoring software. In this figure it is possible to notice the following blocks:

- *SensBox*: The device constantly detects the measurements from the sensors and forwards the data to the Cloud platform. Based on the response received from the platform, the SensBox decides whether to enter in the *On-Alert* mode or not.
- *Receiving*. When the data arrives on the platform through REST services. Data reception can take place via a secure HTTPS connection or, in presence of an Ethernet connection, by forwarding data with a private encryption key via HTTP protocol.
- *Conversion*. It is a set of rules which, based on the device configuration, decide how to convert the received data.
- *Filtering*. It is a set of rules that define any filter criteria to include or discard the detected data.
- *Registration*. It is the passage in which the data is physically saved on the server, within a complex data structure that guarantees high reliability for storage and for accessing the collected information.

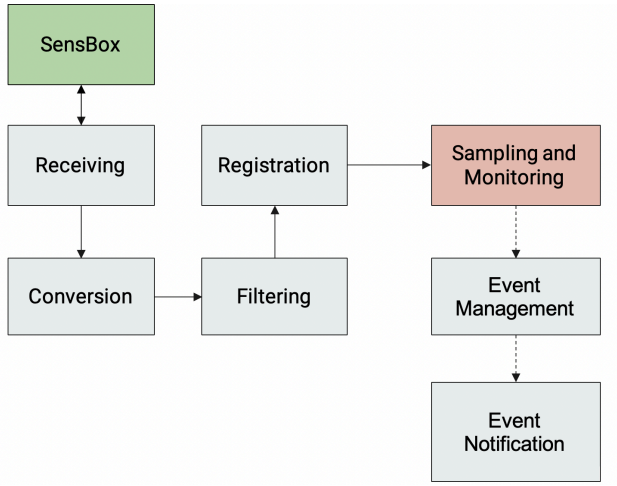


Fig. 2. Monitoring software activities.

- *Sampling and monitoring*. When the system decides if a specific sensor is in alarm or not through appropriate sampling strategies and data analysis that can be defined directly by means of special visual editors.
- *Event Management*. If an alarm occurs, a control system is responsible for storing the details related to that event.
- *Event Notification*. If an alarm occurs, it receives the details of the event and assesses whether it is necessary and with which channels (SMS, E-mail etc.) a notification must be sent to a specific group of people (previously configured).

In addition to receiving data from devices, the system constantly checks the stored data by evaluating the collected information. There are two types of "monitoring" activities:

- **real-time:** whenever information is received from one or more sensors, the data are processed immediately, and any anomalous values are detected;
- **sampling:** the processing activities are based on historical data series or simply on a larger range of information than the single telemetry data.

In this phase an autonomous component called *Monitor Engine* comes in. It is responsible for the evaluation of the sampled data. This control method is more reliable because it bases the generation of any alarms on the analysis of many data, avoiding false positives. This tool works behind the scenes carrying out continuous checks through appropriate scheduling rules deduced on the basis of the configuration of the managed components.

The platform autonomously manages the system which, in general, does not require human intervention except in the setup of new monitored environments. However, a complete interface allows the management of every single aspect of the system. All graphic interfaces are designed to be used both from Desktop and Smartphone.

IV. Validation

In the functional validation some screenshots related to the installer mobile application and other related to the front-end application for the configuration and management of the SensBoxes are presented.

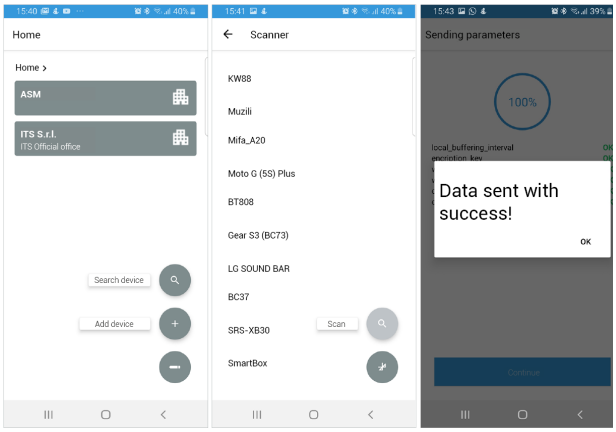
Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on January 07,2026 at 09:45:34 UTC from IEEE Xplore. Restrictions apply.

The installer app allows the configuration of the SensBoxes. The technical operator can navigate within the mapped locations (Fig. 3.a).

After identifying the location in which to install the device, the operator connects the device to an electrical source (electrical socket or PoE) and starts the Bluetooth scan to obtain a direct connection with the device (Fig. 3.b).

Once the Bluetooth connection is successful, the installer can configure the device and save the configuration (Fig.3.c).

The configuration panel of the installed SensBoxes can be consulted from the front-end application. For each SensBox it is possible to visualize its details and the list of connected sensors. Finally, using the monitoring tools, it is possible to analyse the data received in real time (Fig. 4).



a) b) c)

Fig. 3. Some screenshots related to the installer mobile application. List of the mapped locations (a), Bluetooth scanning and connection (b), and configuration saving (c).

V. Discusson

The system just described responds to the need of the health system to treat and stock in a safe and effective way all product categories, cells, tissues and substances used. substances and First of all biological samples (blood, tissues, cells, etc.), unique and not always replicable, for which it is essential to keep their conditions unchanged during sampling and treatment phases; then medical devices, drugs and vaccines. For these substances a certification of the temperatures and deviations from the ranges for conservation is required.

The use of such a system requires a processes reorganization in the healthcare structures and a formalization of the procedures for the management of biological samples, drugs, vaccines, especially in case of emergency and when a malfunction of the refrigeration systems occurs.

The great advantage of the proposed system relies on is its ease of use, its versatility and its minimal impact when introduced in existing healthcare processes. Furthermore, it allows the use of a unique management system for all refrigeration systems used in a company, wherever they are located, independently from their manufacturer. The developed software, with a simple interface that can be used from both desktop and smartphone, allows the users to be profiled differently according to their role into the company, to profile different users with different authorization based on the corporate role and to access the software by logging into the system with the same credentials they use already used to access the corporate domain (single sign-on). The designed system allows to integrate and control, with appropriate precautions, also the devices already present in the company; the only requirement is the presence of a drill and a channel for introducing the probes in the area to be monitored. The cost of the system is lower than similar commercial systems for which many features are optional and not present on all models, especially for small volumes. By using what other producers offer, it might be necessary to substitute most obsolete refrigerator systems. The proposed system gives the possibility to eliminate also allows to reduce costs related to the use of obsolete archiving systems (graphic recorders, diskettes, USB drivers.) and thus also the related management costs. Furthermore, also the costs for the staff dedicated to

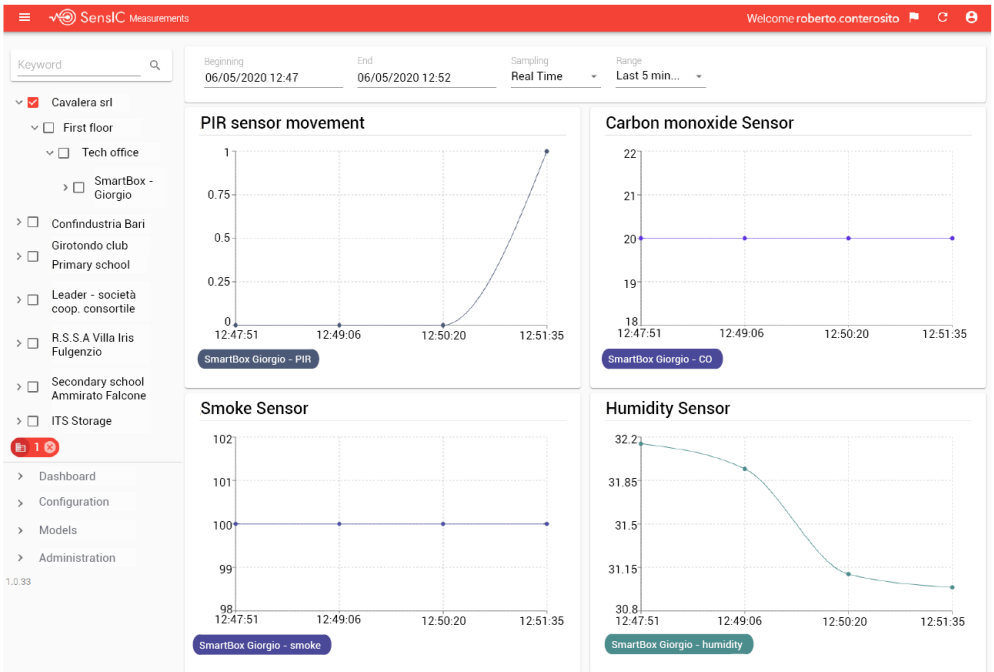


Fig. 4. An example of the available monitoring tools with details on data received in real time.

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on January 07,2026 at 09:45:34 UTC from IEEE Xplore. Restrictions apply.

control and maintenance are reduced as well as costs for the disposal of substances not properly treated.

In Table I, the cost analysis of the entire lifecycle of the project (estimated life 15 years) is reported.

SensIC not only ensures compliance with current legislation and guidelines relating to the conservation of drugs, vaccines, medical devices and biological samples but it also implements a high-quality system.

VI. CONCLUSIONS

In this paper, a complete hardware and software system able to constantly monitor refrigerator systems that transport temperature sensitive biological samples has been proposed. The proposed system provides several users with both real time monitoring dashboards and historical data series related to detected parameters.

Furthermore, it offers an immediate alarm system to manage the risk associated with the malfunctioning of refrigeration systems. In this way, health personnel can intervene promptly in order to prevent the deterioration of the substances transported. Among all the advantages that a system of this type brings to the health system, the huge cost savings due to staff dedicated to the control and maintenance of refrigeration systems and costs related to the disposal of deteriorated substances is certainly not negligible. Compared to the commercial systems currently used in the health sector, the proposed system represents a complete solution capable of satisfying all the needs expressed by the health system for the management and transport of drugs, vaccines, medical devices and biological samples. Further analysis could be performed in the future to focus on security, with the aim to prevent thefts of biological samples, drugs, vaccines and costly medical devices as well on the usage of the system in case of very low temperatures (certification of cryopreservation).

TABLE I. COST COMPARISON (LIFECYCLE 15 YEARS, COMPONENTS DETERIORATION COSTS NOT TAKEN INTO ACCOUNT)

Category	Description	Traditional System	Other systems	SensIC
Temperature monitoring	Recording diagram	€ 600,00	€ 0,00	€ 0,00
Temperature monitoring	Diskettes	€ 450,00	€ 0,00	€ 0,00
Temperature monitoring	USB-pen/batteries	€ 1.800,00	€ 0,00	€ 0,00
Temperature monitoring	Probe	€ 0,00	€ 100,00	€ 60,00
Temperature monitoring	Datalogger / Signal Concentrator	€ 0,00	€ 300,00	€ 225,00
Temperature monitoring	Lan Receiver	€ 0,00	€ 900,00	€ 0,00
Temperature monitoring	General costs (employees)	€ 5.400,00	€ 1.350,00	€ 1.350,00
Temperature monitoring	Maintenance	€ 420,00	€ 910,00	€ 798,00
Alarm monitoring	Alarm accessories	€ 100,00	€ 600,00	€ 0,00
Alarm monitoring	General costs (employees)	€ 5.400,00	€ 1.350,00	€ 1.350,00
Alarm monitoring	Maintenance	€ 70,00	€ 420,00	€ 0,00
Alarm and temperature monitoring	Software	€ 0,00	€ 2.000,00	€ 0,00
Alarm and temperature monitoring	Maintenance	€ 0,00	€ 1.400,00	€ 0,00
	Total	**€ 14.240,00**	**€ 9.330,00**	**€ 3.783,00**

ACKNOWLEDGMENT

Part of the work in this paper was carried out within the SAFETY (Smart Aid System for Fire and Danger situations based on IOT technology) project. SAFETY is a research project with code 6RPCRL1. ("Intervento cofinanziato nell'ambito del POR Puglia FESR-FSE 2014-2020 - Asse prioritario 1 - Ricerca, sviluppo tecnologico, innovazione - Azione 1.4.b - BANDO INNOLABS – SOSTEGNO ALLA CREAZIONE DI SOLUZIONI INNOVATIVE FINALIZZATE A SPECIFICI PROBLEMI DI RILEVANZA SOCIALE). Public Notice and subsequent reissues. AD n. 144/13 of 08/02/2017- Section DD Research and Innovation Capacity 'Institutional n. 37 of 28 March 2017.

REFERENCES

[1] D. Alessandrelli, L. Mainetti, L. Patrono, G. Pellerano, M. Petracca, M. L. Stefanizzi, "Implementation and validation of an energy-efficient MAC scheduler for WSNs by a test bed approach," in SoftCOM 2012, 20th International Conference on Software, Telecommunications and Computer Networks, Splot, Croatia, 2012.

[2] L. Catarinucci, S. Guglielmi, L. Mainetti, V. Mighali, L. Patrono, M. L. Stefanizzi, L. Tarricone, "An Energy-Efficient MAC Scheduler based on a Switched-Beam Antenna for Wireless Sensor Networks," Journal of Communications Software and Systems (JCOMSS), 9(2), pp. 117-127, 2013.

[3] D. Alessandrelli, L. Mainetti, L. Patrono, G. Pellerano, M. Petracca, M. L. Stefanizzi, "Performance Evaluation of an Energy-Efficient MAC Scheduler by using a Test Bed Approach," Journal of Communications Software and Systems (JCOMSS), 9(1), pp. 84-96, 2013.

[4] ISPE, Good Practice Guide: Cold Chain Management, ISBN-13: 978-1936379064: ISBN-10: 1936379066, May 2011.

[5] Declaration of Storage Conditions for Medicinal Products Particulars and Active Substances. Online: https://www.ema.europa.eu/en/declaration-storage-conditions-medicinal-products-particulars-active-substances-annex (March 2020).

[6] Guide to the quality and safety of tissues and cells for human application, 2nd Edition. Online: https://www.edqm.eu/sites/default/files/foreword_list_of_contents_tissues_cell_guide_2nd_edition_2015.pdf (March 2020).

[7] Guide to the preparation, use and quality assurance of blood components, 19th Edition. Online: https://www.edqm.eu/sites/default/files/list_of_contents_19th_ed-blood-quality.pdf (March 2020).

[8] European regulation (EU) 2017/745. Online: https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32017R0745 (March 2020)

[9] Discovery kit for IoT node. Online: https://www.st.com/content/st_com/en/products/evaluation-tools/product-evaluation-tools/mcu-mpu-eval-tools/stm32-mcu-mpu-eval-tools/stm32-discovery-kits/b-l475e-iot01a.html (March 2020).

[10] STM32L4 Series. Online: https://www.st.com/en/microcontrollers-microprocessors/stm32l4-series.html (march 2020).

[11] DS18B20 Datasheet. Online: https://datasheets.maximintegrated.com/en/ds/DS18B20.pdf (March 2020).

[12] PT-100 Platinum RTDs Datasheet. Online: http://irtfweb.ifa.hawaii.edu/~iqup/domeenv/PDF/pt100plat.pdf (March 2020).

[13] MariaDB. Online: https://mariadb.com/ (March 2020).

[14] Cake PHP framework. Online: https://cakephp.org/ (March 2020).

[15] Hetzner. Online: https://www.hetzner.com/ (March 2020).

[16] ReactJS. Online: https://reactjs.org/ (March 2020).

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on January 07,2026 at 09:45:34 UTC from IEEE Xplore. Restrictions apply.